# POKHARA UNIVERSITY

Level: Bachelor | Semester : Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Object Oriented Programming in C++ | Pass Marks: 45
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

1. a) Explain non-linear behaviour of complexity. How is OOP different from POP? Justify with points. 7

**OR**

Define Responsibility Driven Design. Demonstrate the significance of CRC in OOP with an example.

b) What shortcomings of structure are solved by classes? Explain different types of constructors available in C++ with their use. 8

2. a) Does the friend function breach Data hiding? Justify. Write a program to swap the values of variables of two classes using friend function. 8

b) How is DMA better than SMA? Justify your answer. Explain how new and delete operators are used in DMA. 7

3. a) Define static member. Write a program to illustrate the use of static members and static member function. 7

b) How inheritance increases the reusability of codes? Explain different types of inheritance with suitable diagram and syntax. 8

4. a) What is diamond problem in inheritance? Explain how this problem can be solved with suitable programming example. 7

b) Define two ways of converting one user defined data type to another user defined type. Write a program that convert kilogram into gram using user define to user define data conversion. (1kg=1000gm). 8

**OR**

WAP to find the sum of two 3×3 matrix by overloading binary operator +.

5. a) What do you mean by virtual function? Write a program to show the use of virtual function to achieve runtime polymorphism. 7



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b) Why is generic programming beneficial? Explain template function and class template with syntax. 8

6. a) Define Exception handling. Does C++ support multiple catch? Illustrate with suitable programming example. 7

b) Write a program that stores object of student class (assume data members are roll, name and university_ name) into a file and read values from the file and display the data in console. 8

7. Write short notes on: (Any two) 2×5

a) STL

b) Use of this pointer

c) OOAD

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